



The Red Sea :

**Anne Vanhove
Océane Rosenberg
Ylenia Reuter**

**Description
Plate tectonics
Sedimentology
Tracers
Conclusion**

Geographical location



Map data ©2023 Google

- Bordered by seven countries
- Surface area of approximately 438,000 km²
- Elongated sea: 2,250 km long and 355 km wide
- Located between the African and Arabian plates
- Sea surface temperature ranging from 22°C to 32°C
- Arid climate with low precipitation
- Microtidal regime
- Salinity of 41 g/L

Topography and wind patterns :

Topography :

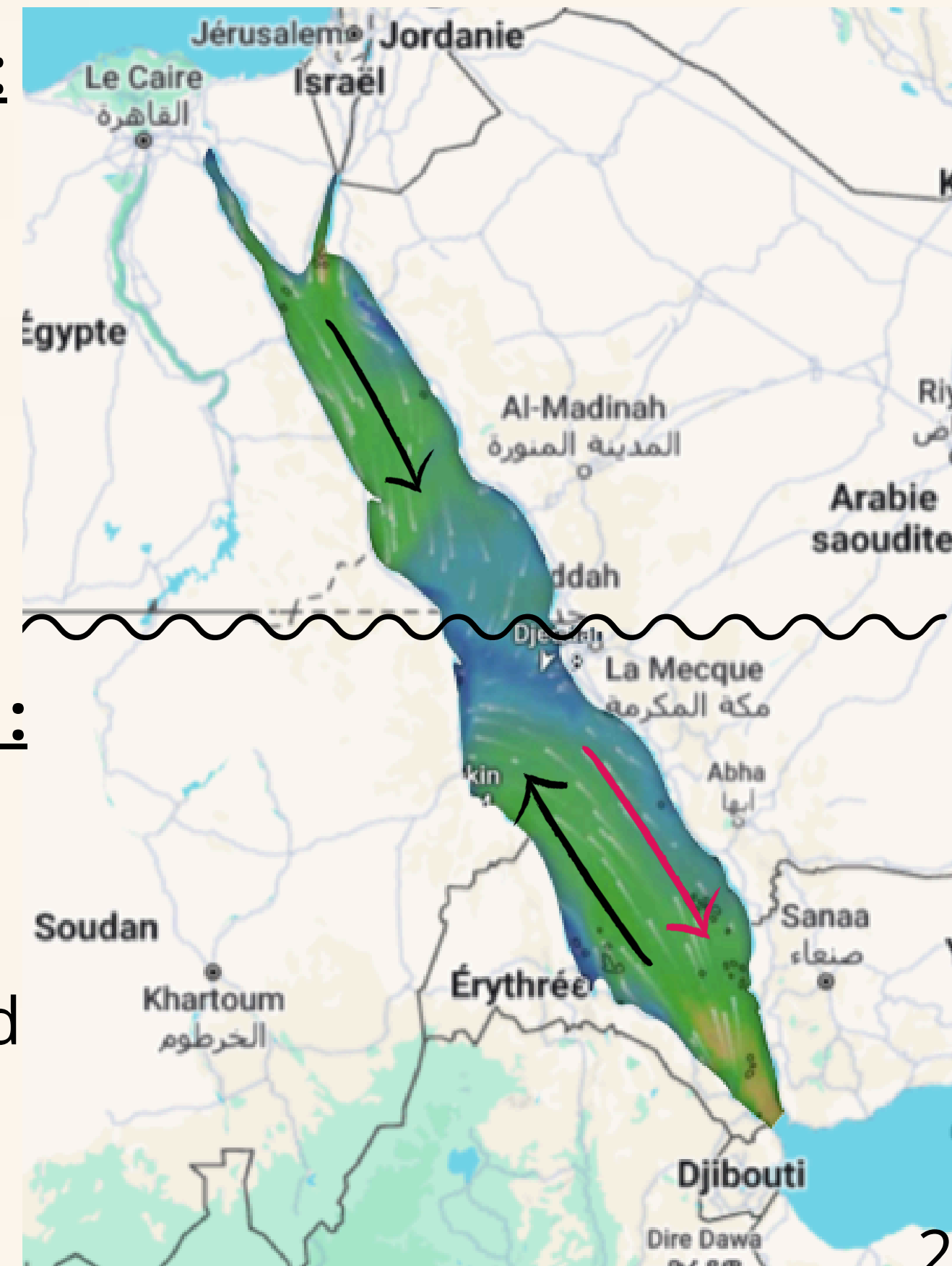
Winds :

Northern part :

Annual
northwesterly
winds

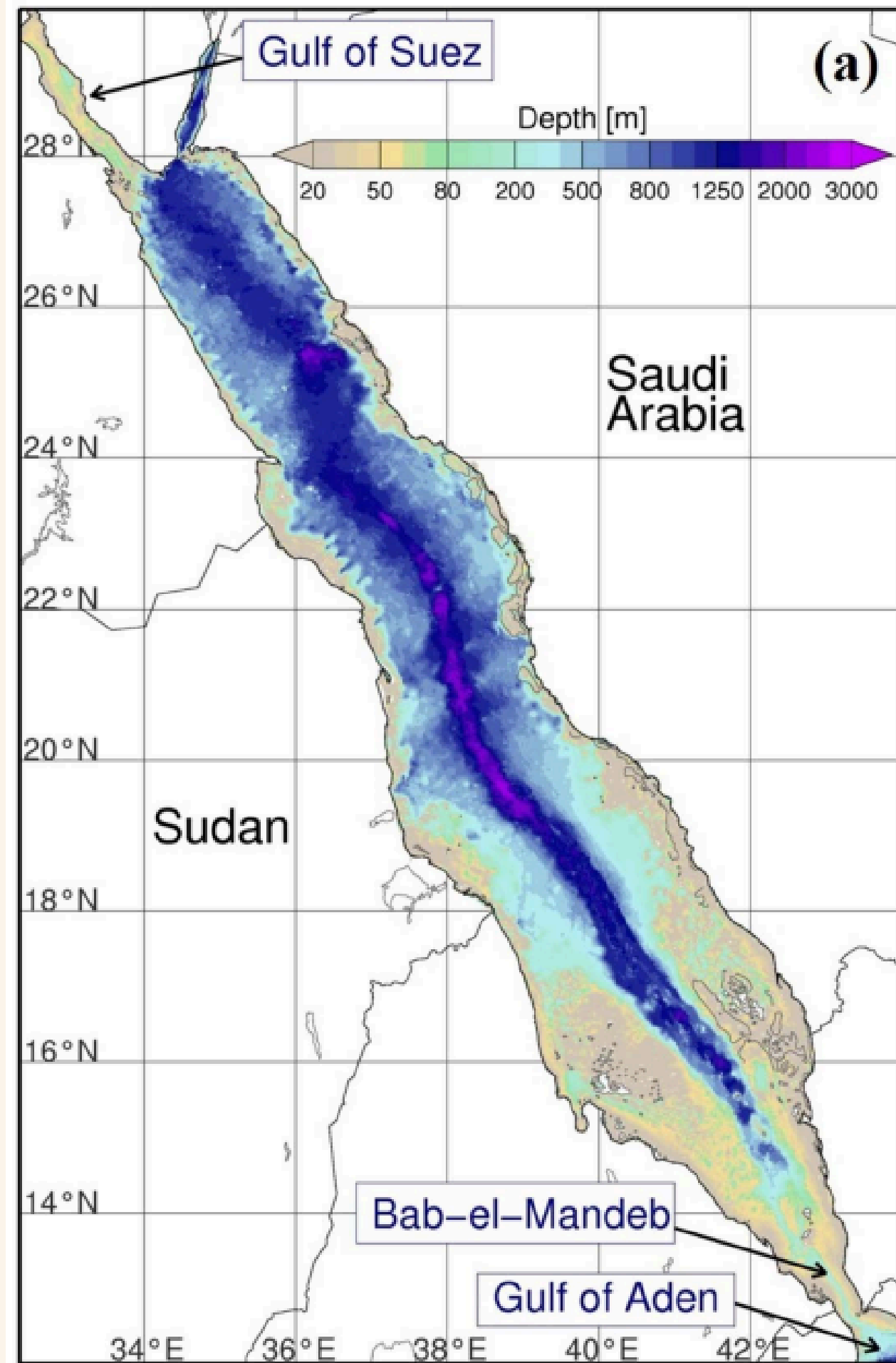
Southern part :

- Winter : SO wind
- Summer: NO wind

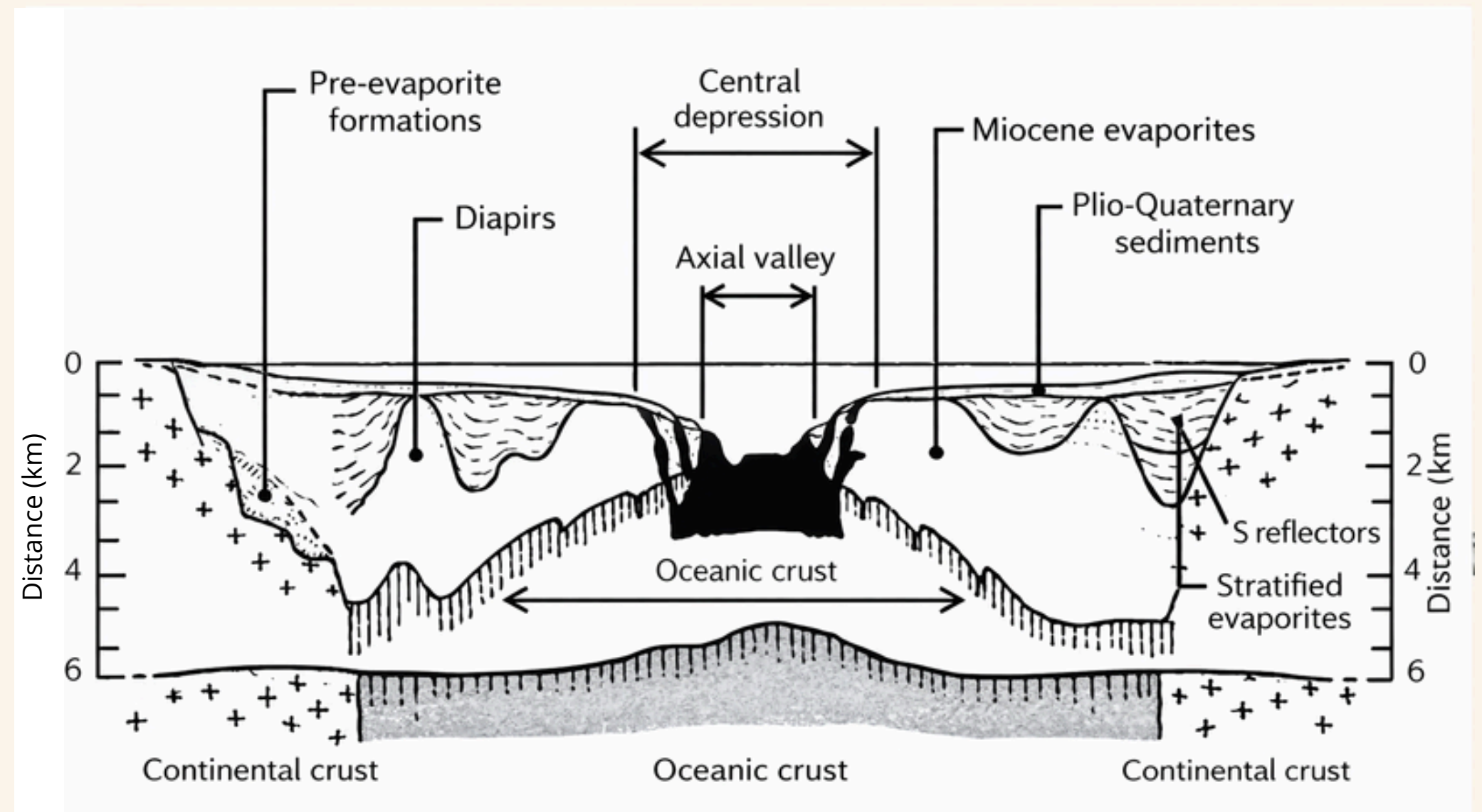


Bathymetry.

- Maximum depth: 2,600 meters
- Average depth: 500 meters
- Extensive continental shelf (Béthoux, 1986)
- Different north-south plate velocities



(Raitsos et al., 2013)



Birtin et al., 1979 (cited by Pautot, 1983).

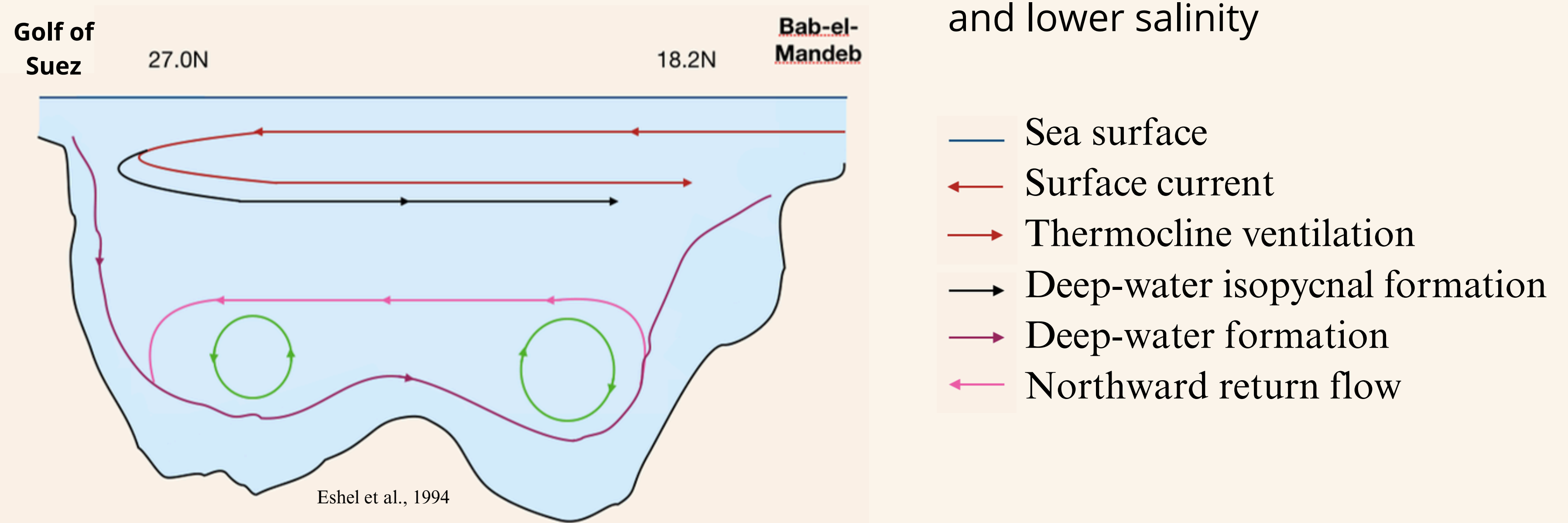
Ocean currents :

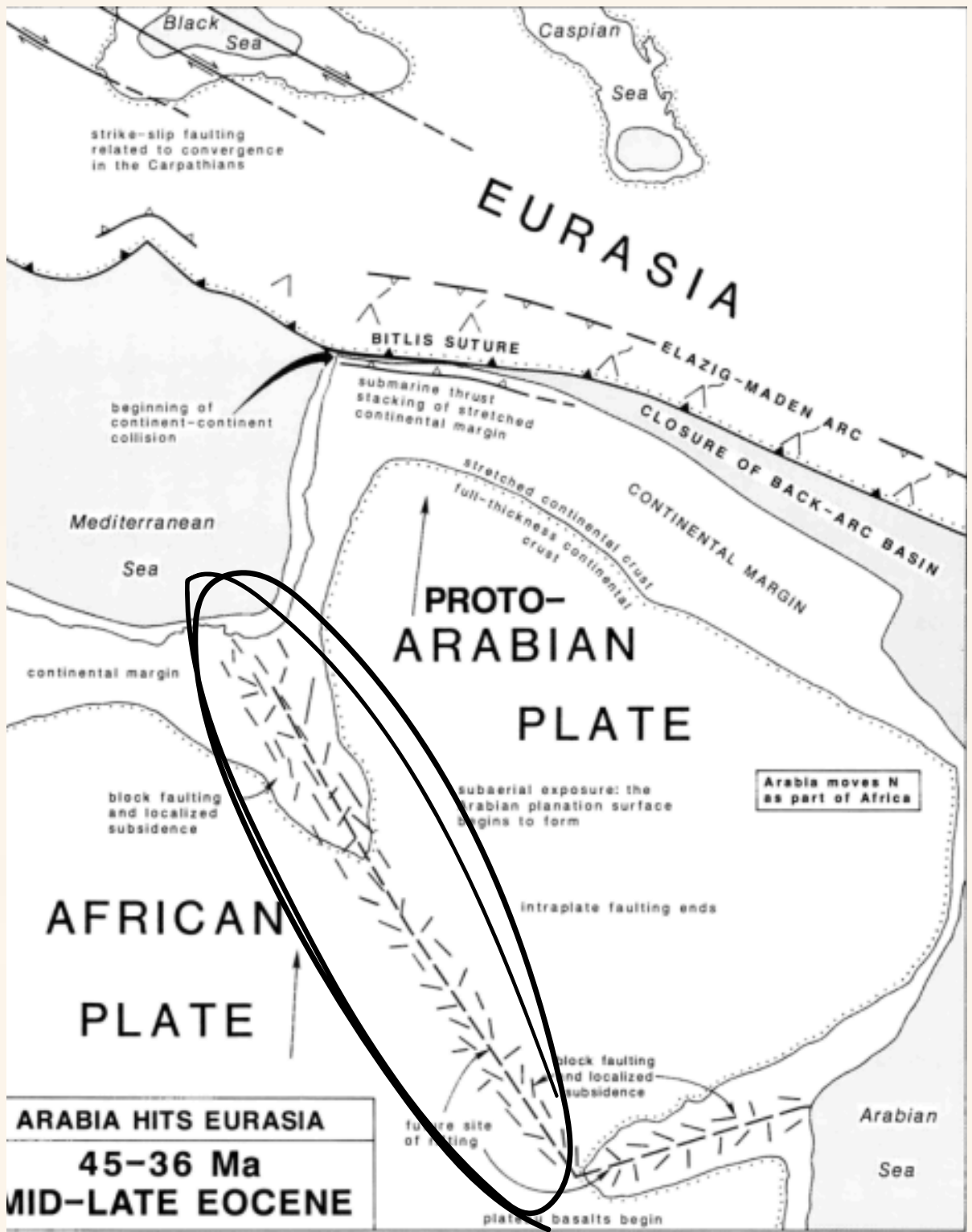
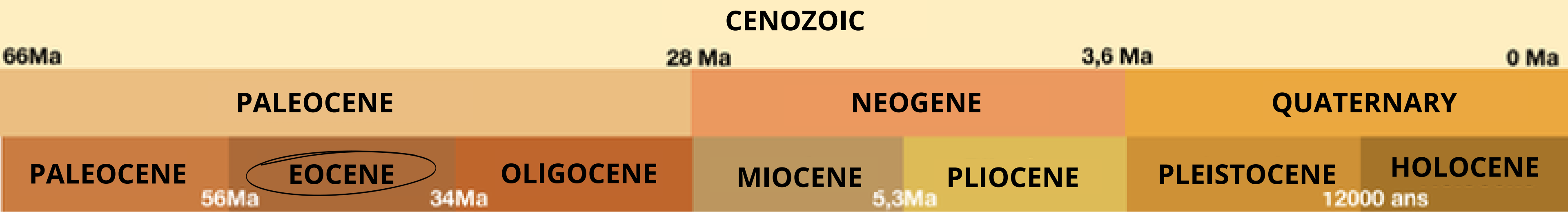
Mousson d'hiver:

- Vents N-O et S-E + baisse P_{atm}
- Création d'un courant de surface
- Eaux intermédiaires et profondes sens opposé, (salinité élevée)

Summer monsoon:

- Reversed wind patterns + increase in atmospheric pressure
- Reversal of surface currents
- Intermediate waters from the Gulf of Aden sink due to lower temperature and lower salinity

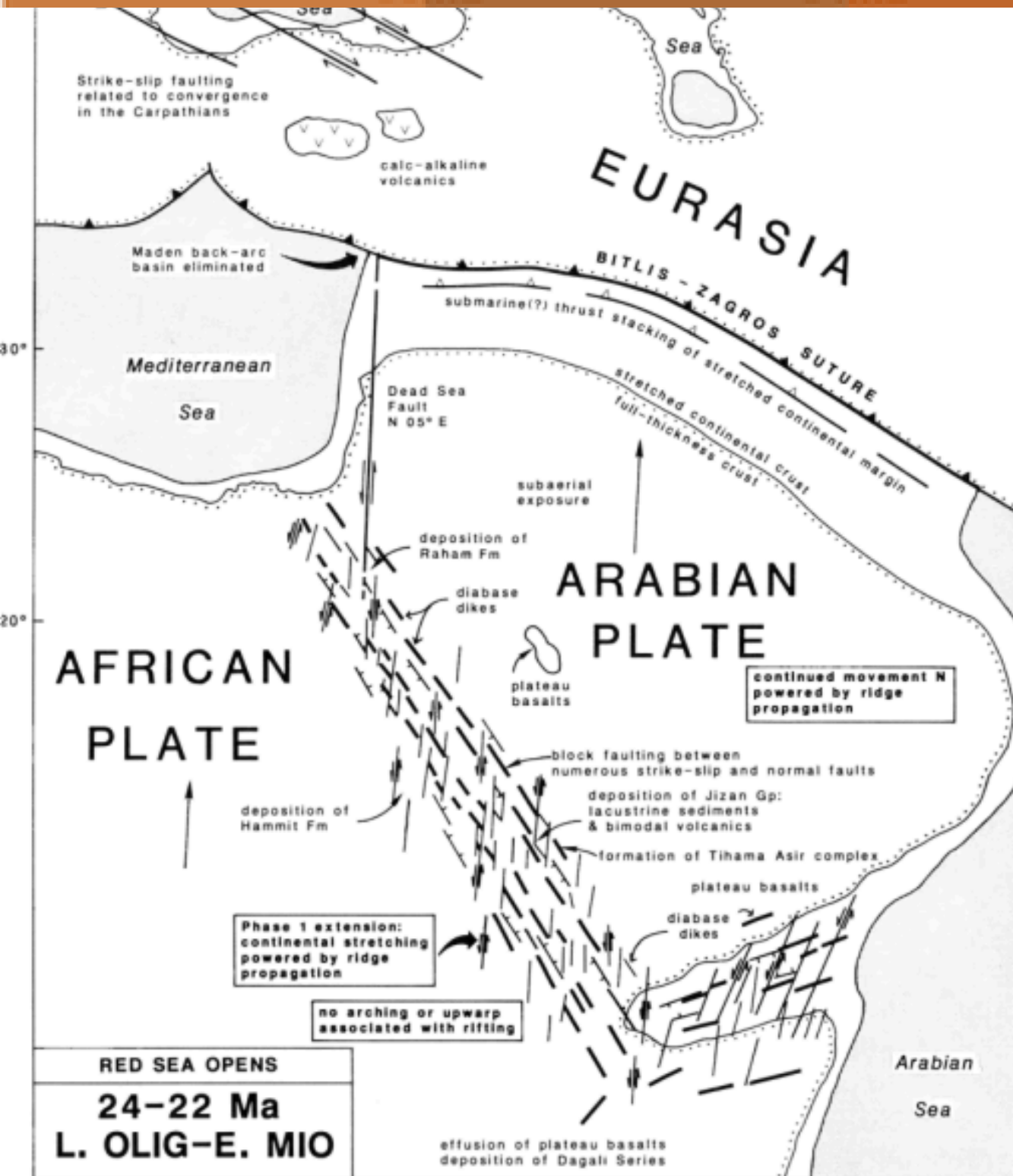
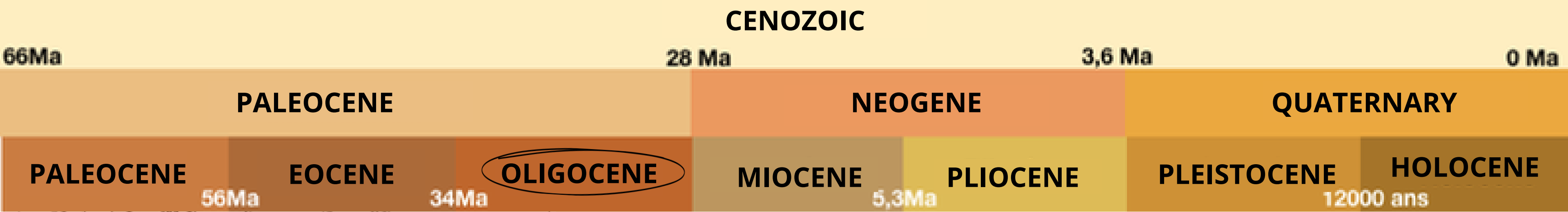




- Divergence of the African and Arabian tectonic plates
- Around 30 million years ago: the Arabian Plate separated from the African Plate
- This rifting led to the formation of the Red Sea

Late Eocene → Initial stage: fault blocks developed along the Red Sea

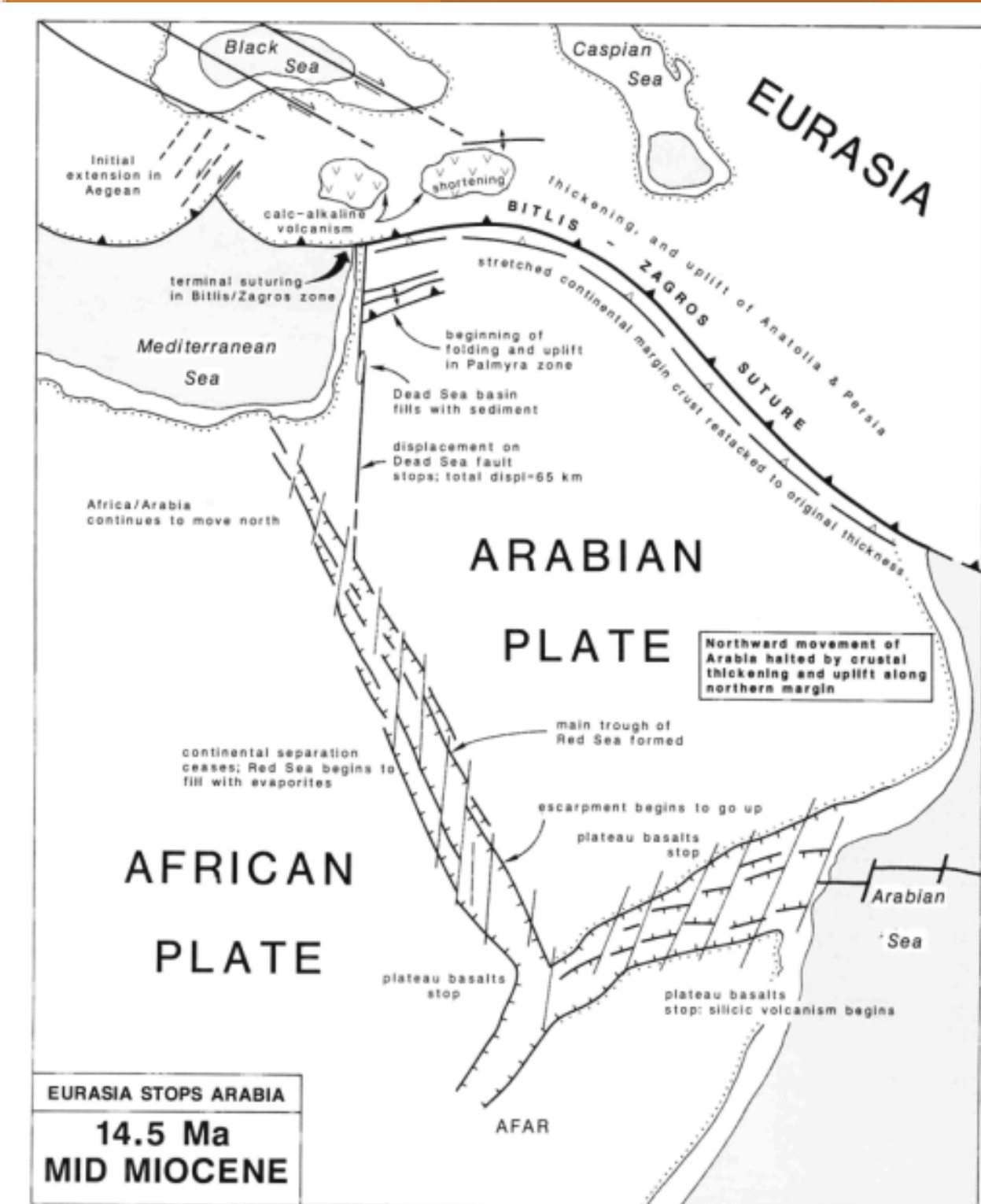
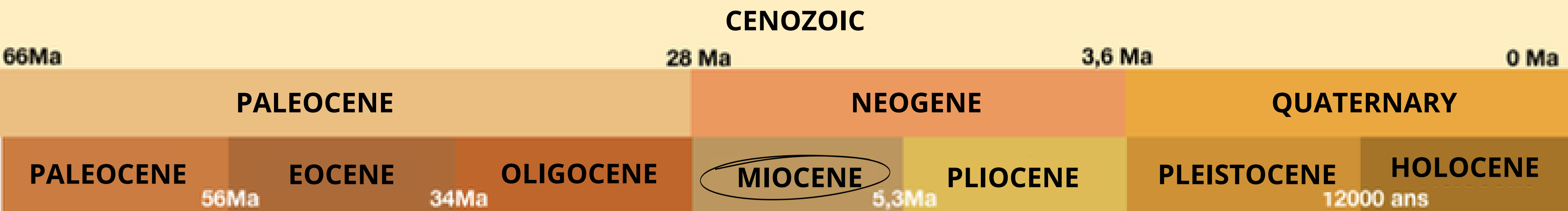
2. Tectonic reconstruction for the mid to late Eocene illustrating the initial suturing between Arab Eurasia.



Oligocene → (basin formation):

- Onset of continental rift volcanism
- → Opening and basaltic intrusions
- Significant extension
- Formation of faults along numerous strike-slip and normal faults
- Probable connection with the Indian Ocean

Fig. 3. Tectonic reconstruction for the late Oligocene to early Miocene illustrating the opening of the Red Sea. Ali et al., 2023



Early Miocene:

- Probable connection with the Mediterranean Sea
- First reefal developments

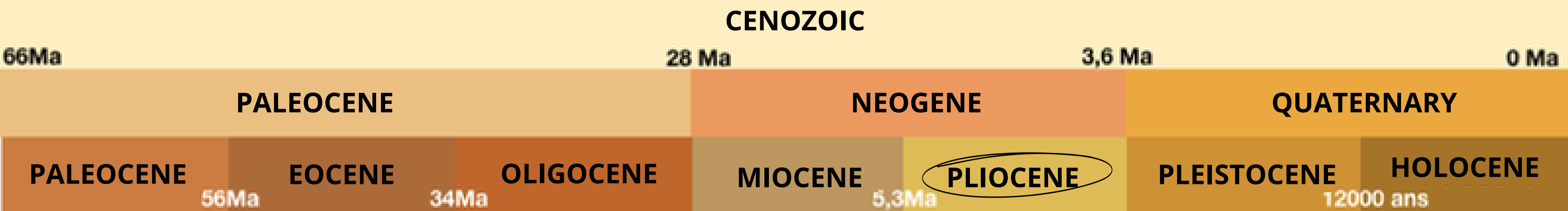
Middle Miocene:

- Salt accumulation and isostatic uplift of basin margins

Late Miocene:

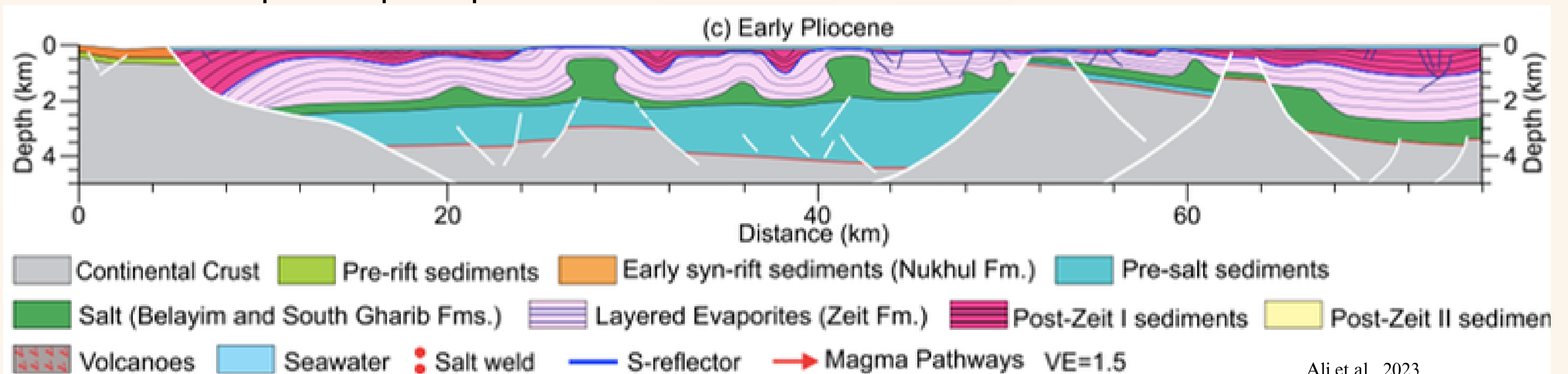
- Intense evaporation → gypsum formation, evaporite deposits, and decline in biodiversity

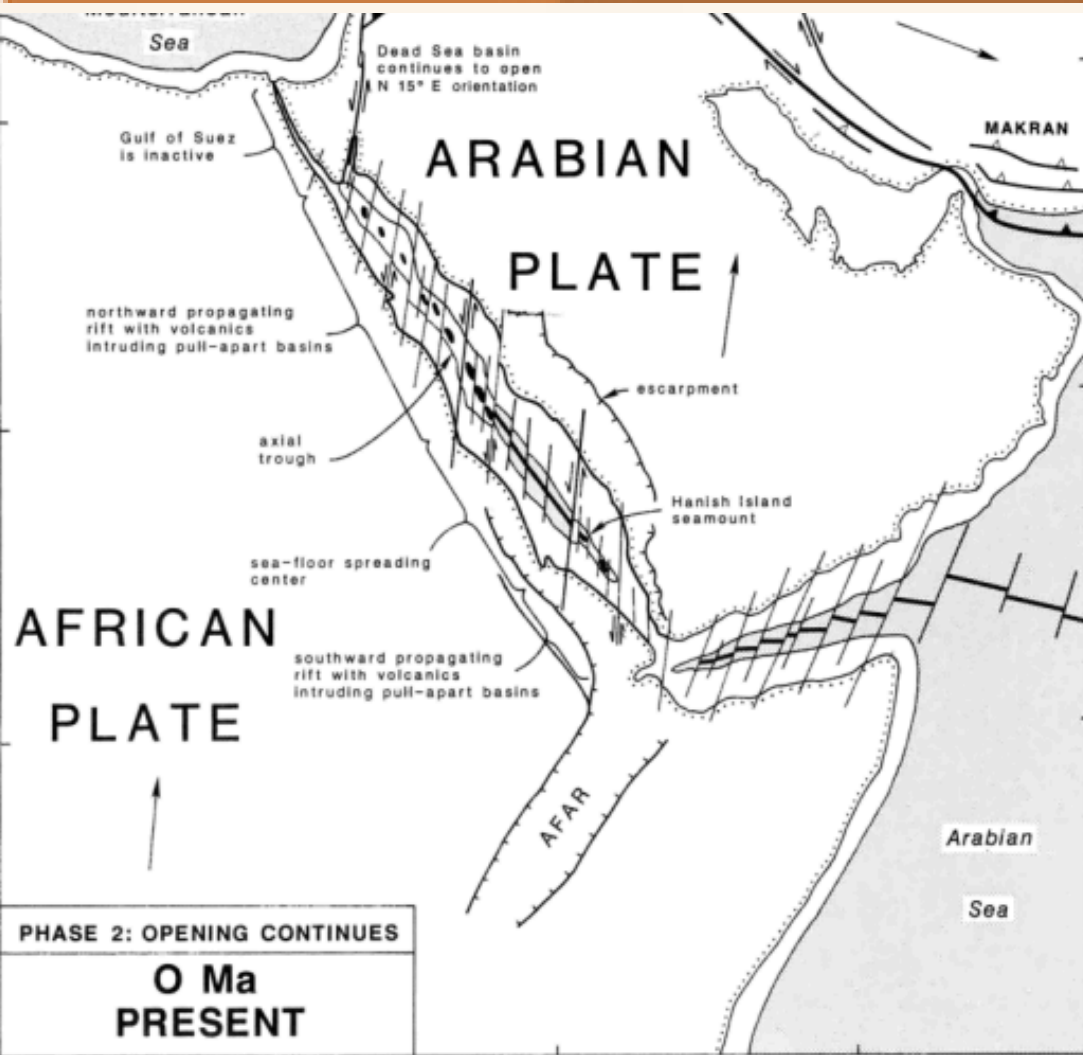
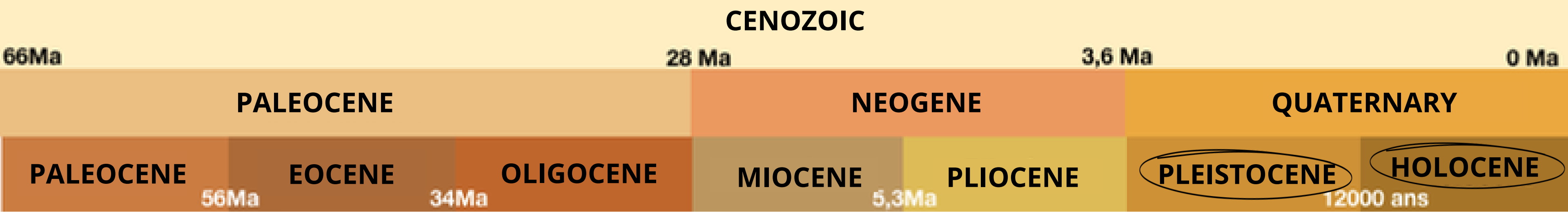
Fig. 4. Tectonic reconstruction for the medial Miocene illustrating terminal suturing between Arabia and Eurasia and the temporary halting of Arabia's independent northward motion relative to Africa.



Pliocene :

1. Thinned and fractured continental crust → intrusion of oceanic basalts
2. Basin widening and deepening + seafloor spreading
3. Connection with the Indian Ocean and biological recolonization (biodiversity, fossils)
4. End of evaporite precipitation and inflow of water

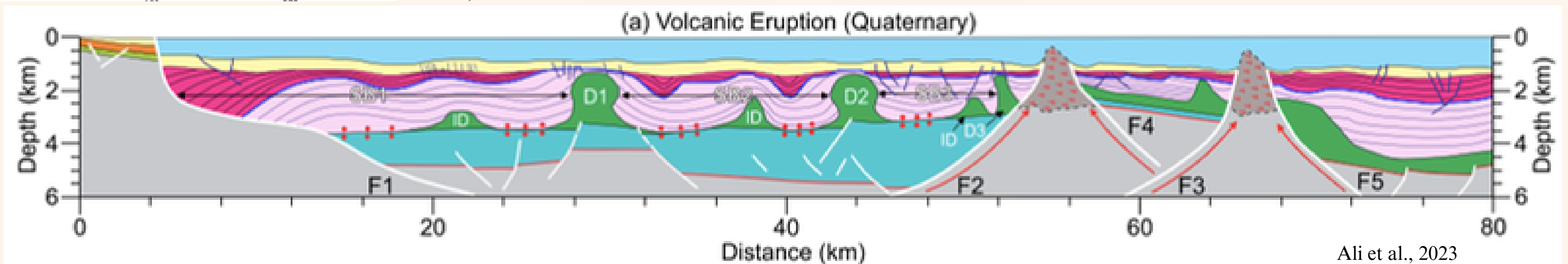




Ali et al., 2023

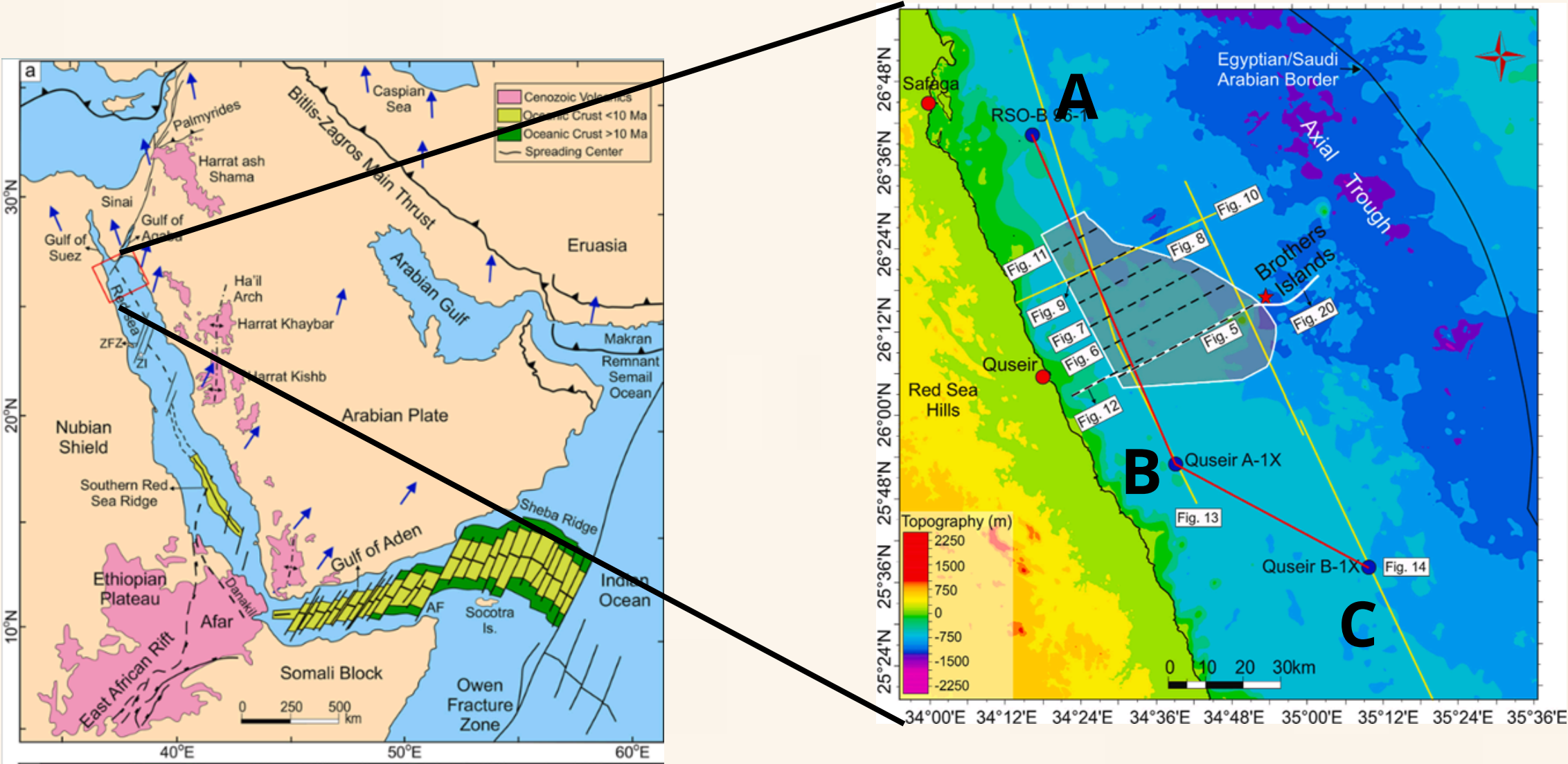
Pleistocene + Holocene :

- The Arabian Plate continues its northward movement
- All plate boundaries remain active
- Seafloor spreading rate of ~20 km per Ma

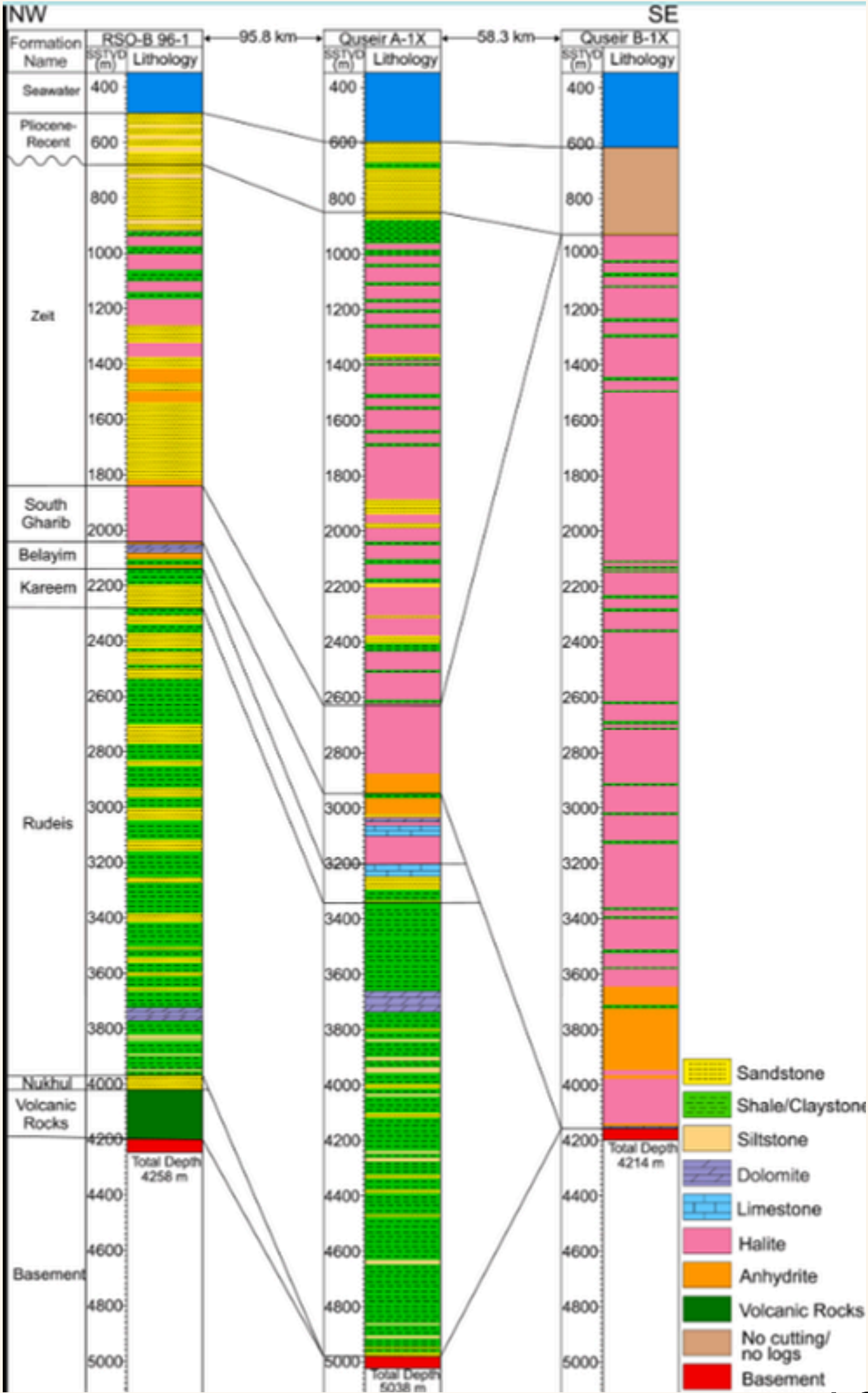


Ali et al., 2023

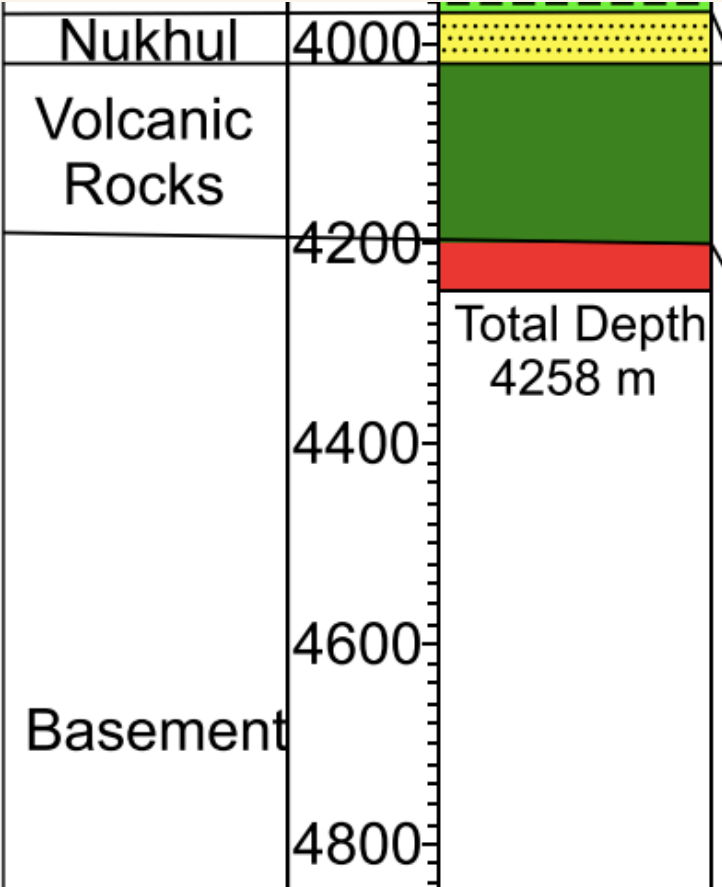
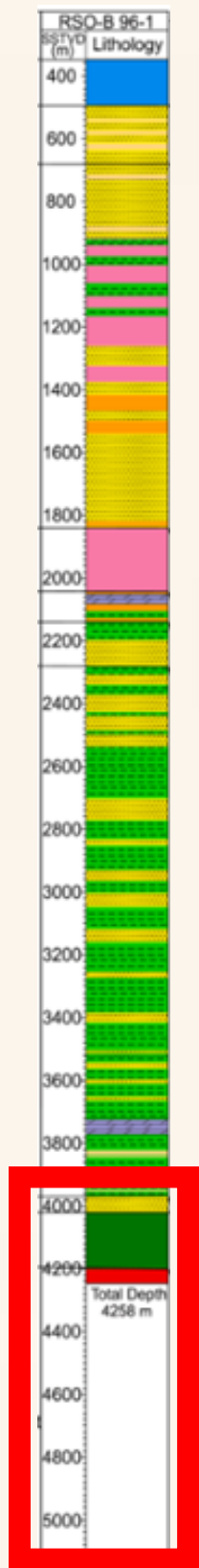
Sedimentology_:



Study area: Northern Egyptian Red Sea

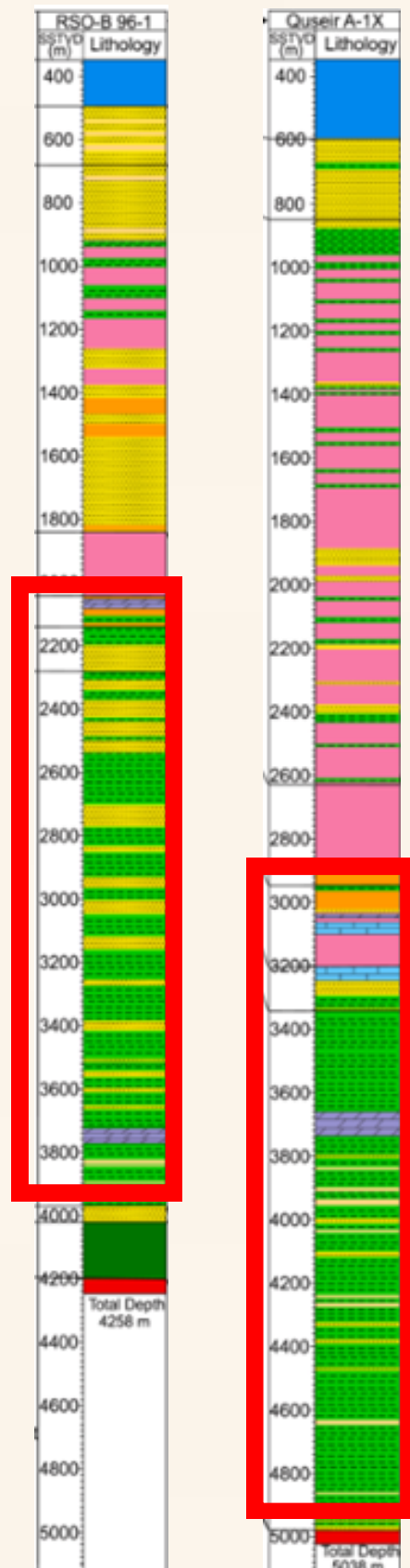


Early miocene :



- Thickness $\approx$ 52 m
- Mainly composed of detrital sedimentary rocks (sandstone)

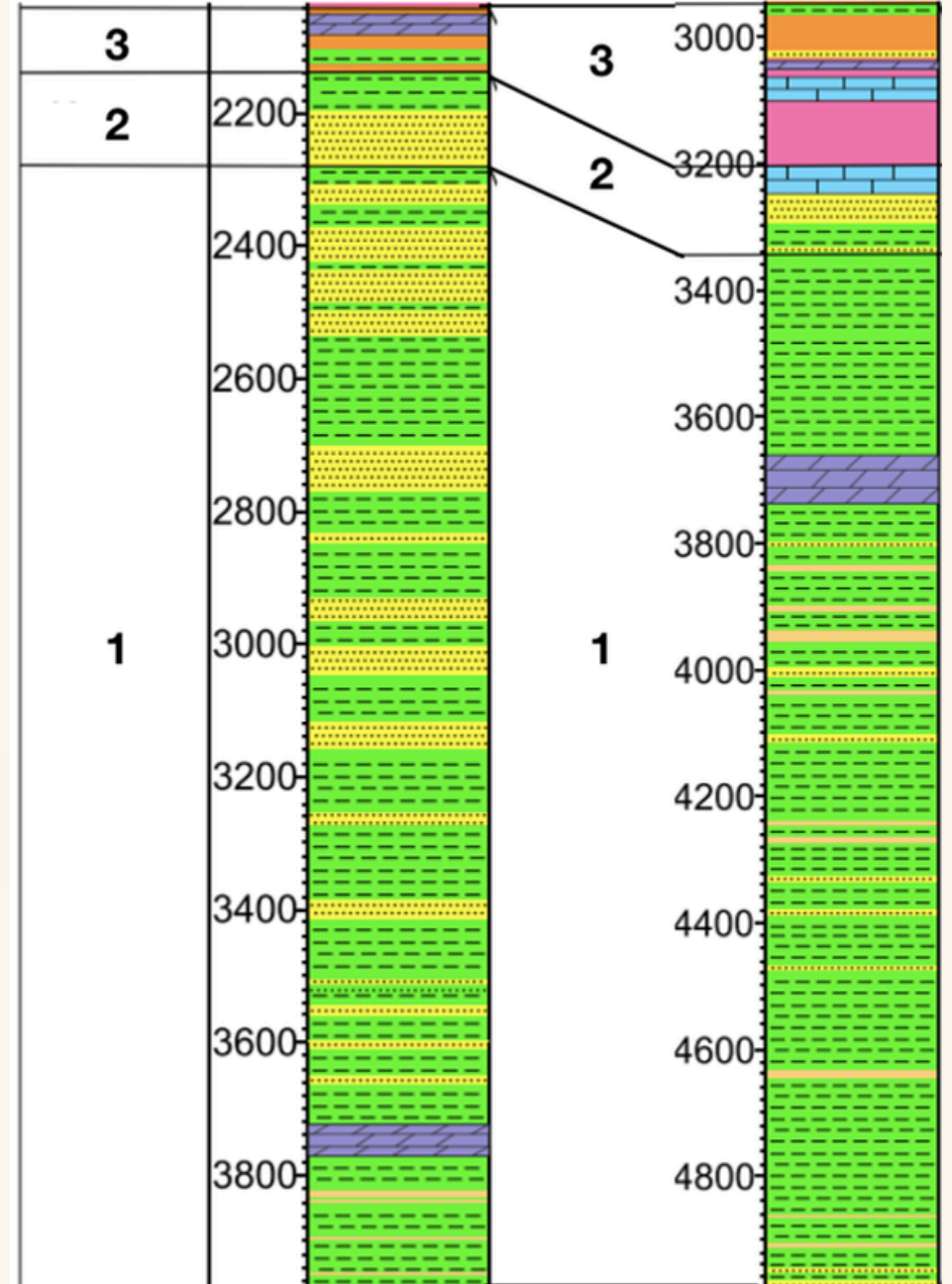
Well A



Well A and B

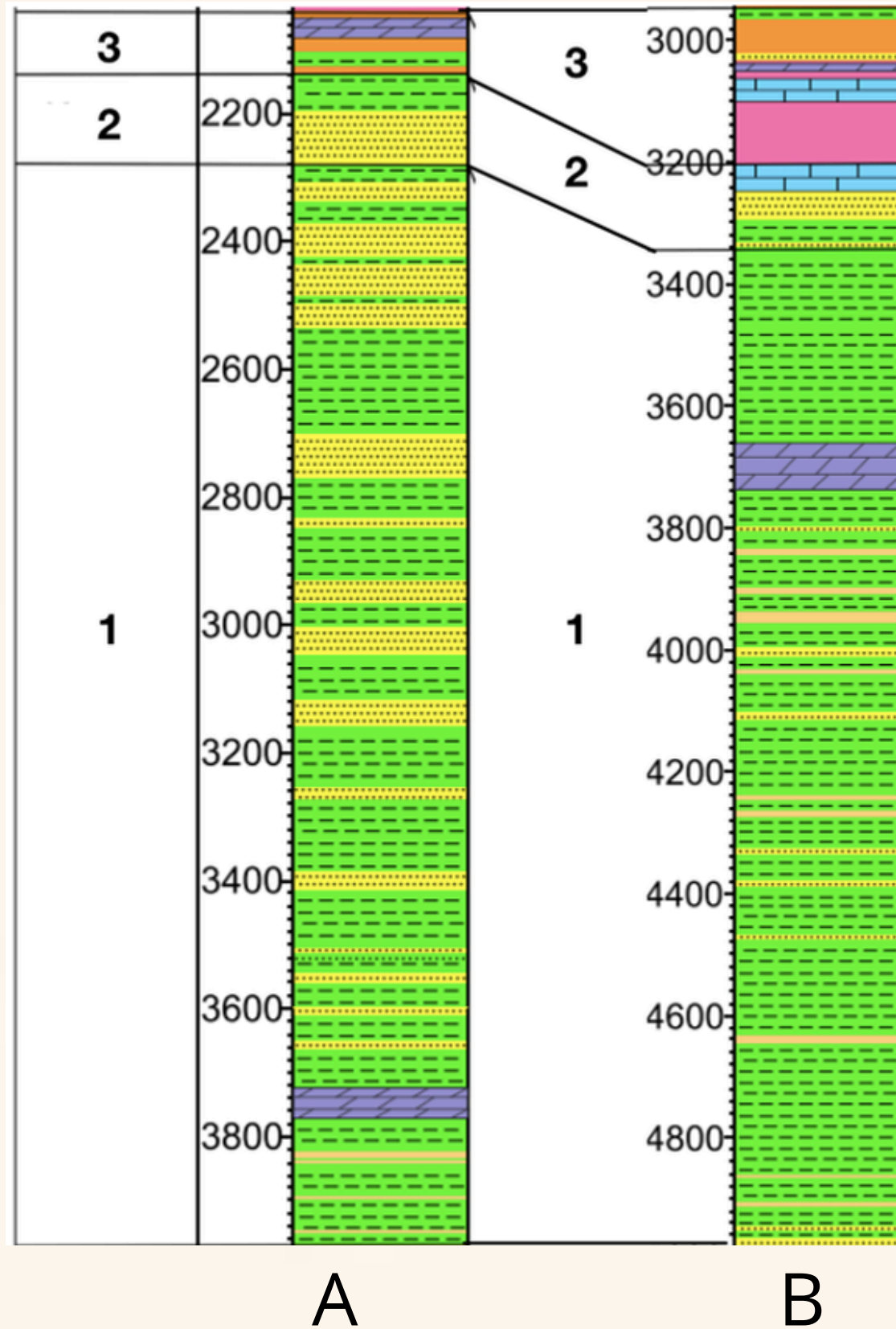
Middle Miocene :

In three stages: layers 1, 2, and 3.



A B

Middle miocene :



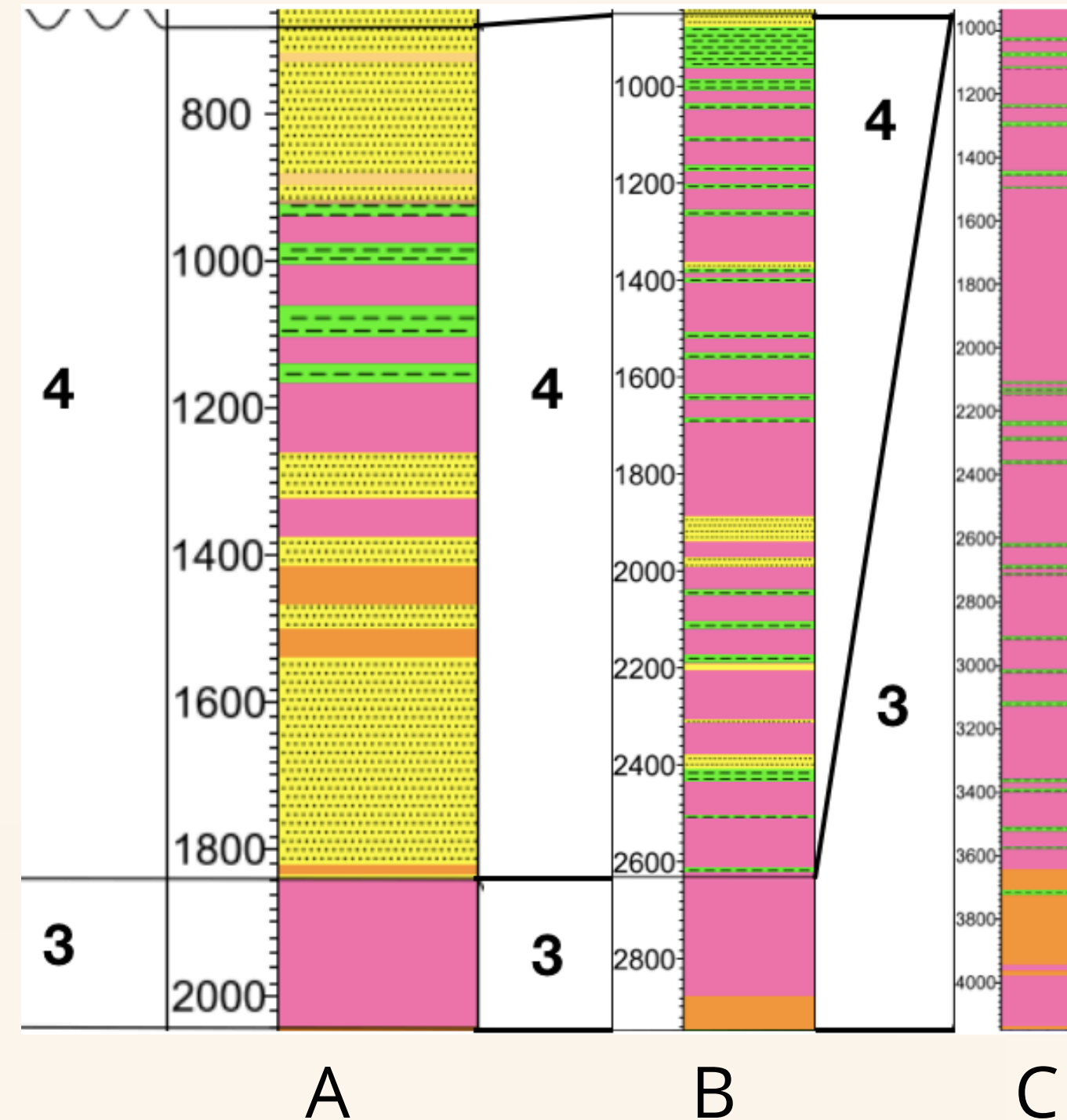
(Ali et al., 2023)



- **Layer 3:** Anhydrite and halite → intense evaporation and very high salinity
- **Layer 2:** Continuation of Layer 1, composed of siliciclastic rocks (clay, siltstone, and sandstone), followed by thin interbedded layers of limestone and dolomite
- **Layer 1:** Clay interbedded with thin layers of sandstone and siltstone → detrital sedimentary rocks



Late miocene :

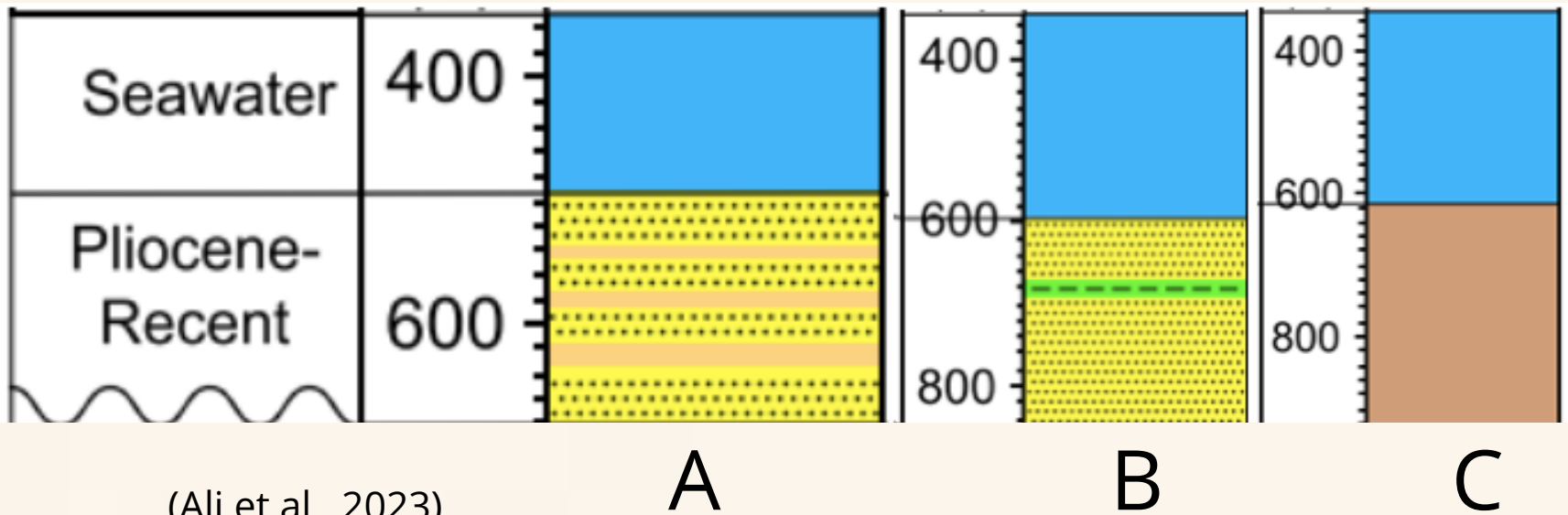


Zone 4: Wells A and B composed of halite and anhydrite, with interbedded levels of sandstone, siltstone, and clay

- Zone 3:** Wells A, B, and C composed of a massive salt layer → halite and anhydrite



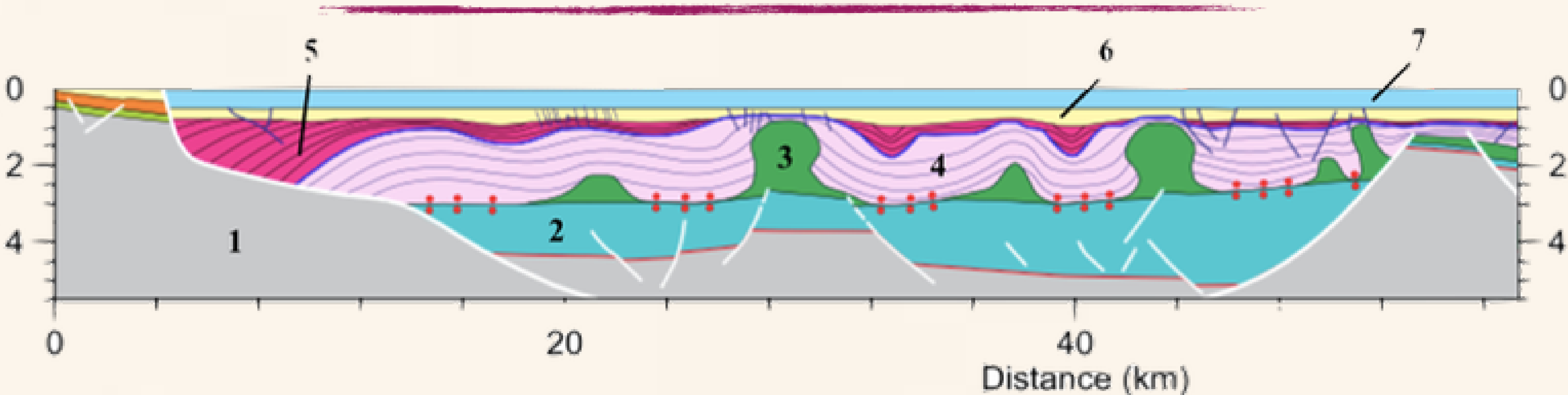
Pliocene-Quaternary:



(Ali et al., 2023)

Composed of sandstone interbedded with thin layers of siltstone and clay

→ Rise in sea level, recovery of biodiversity, and formation of clastic sedimentary rocks



1 – Continental crust; 2 – Pre-salt sediments; 3 – Halite; 4 – Evaporites;
5 and 6 – Recent Pliocene sediments; 7 – Seawater

Paleoceanographic tracers :

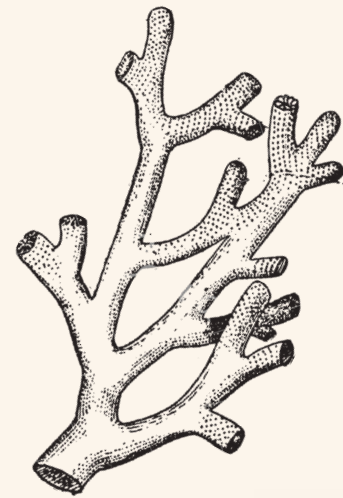
What is the age of the Red Sea reef terraces?

- Method: Uranium-238 / Thorium-230

$$\frac{^{230}\text{Th}}{^{238}\text{U}} = 1 - \exp(-\lambda_{^{230}\text{Th}} t)$$

(Lalou and al., 1979)

- Why this method ?



The skeleton of a coral is made of calcium carbonate.

- Uranium and calcium have a chemical affinity.
- Thorium, in turn, is formed from uranium.
- The $^{230}\text{Th}/^{238}\text{U}$ ratios can therefore provide information about time.

Paleoceanographic tracers :

Materials and methods:

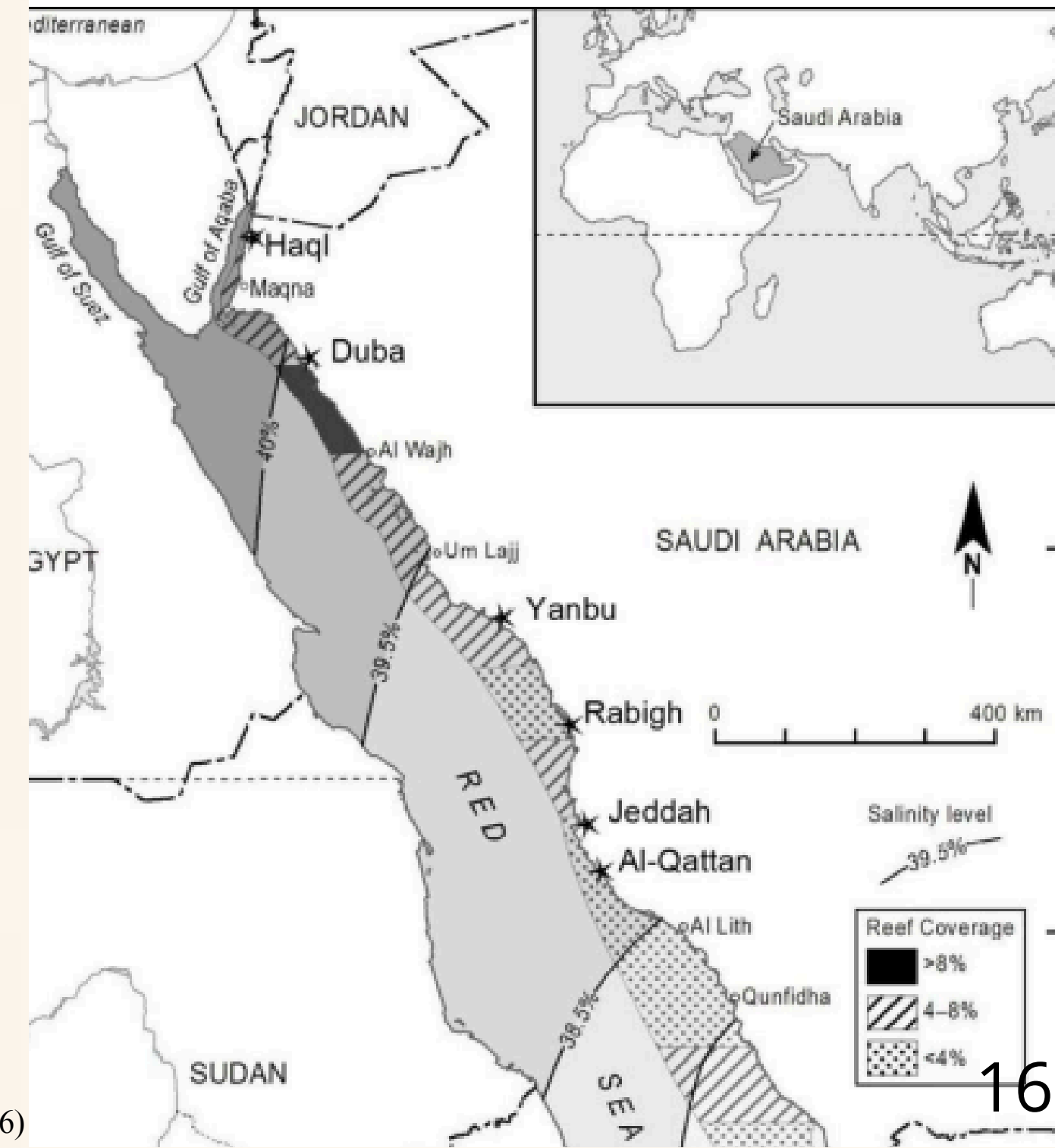
- Location: along the Saudi Arabian coast of the Red Sea
- Technique: samples collected from six different areas

Results:

Age of the upper parts of the terraces: 119 to 125 thousand years.

→ Formation of these terraces during the last interglacial period, MIS 5e

→ MIS 5e = a period when global sea level was 6–9 meters above the present level



Conclusion

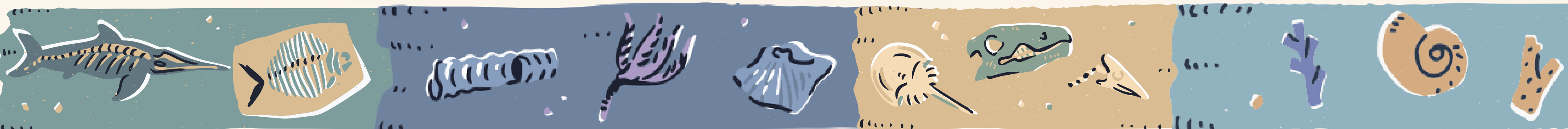
The Red Sea: a basin whose origin is difficult to determine !



- Currents influenced by monsoon winds and salinity
- Basin opening ~30 million years ago
- Divergence of the African and Arabian plates
- Low sediment diversity: evaporites, salt, siliciclastic rocks, limestones
- Coral fossils allow U/Th dating



Thank you for your attention !



Sources :

- Ali, M., Koyi, H., Bosworth, W., Ligi, M., Ball, P. J., Decarlis, A. (2023). Geometry and kinematics of the Middle to Late Miocene salt tectonics, central Egyptian Red Sea margin. *Journal of Structural Geology*, 176.
- Bertin J., Pedeux J.P., Magenham J.C., 1979. Contribution de la géophysique et de l'océanographie physique à la recherche et à l'exploration des boues métallifères de la Mer Rouge. Séminaire International : Ressources minérales sous-marines, Orléans, octobre 1978, Documents BRGM n° 7, 339-365.
- Eshel, Gidon; Cane, Mark A.; Blumenthal, M. Benno (1994). Modes of subsurface, intermediate, and deep water renewal in the Red Sea. *Journal of Geophysical Research*, 99(C8), 15941–. doi:10.1029/94jc01131
- Hempton, M. R. (1987). Constraints on Arabian plate motion and extensional history of the Red Sea. *Tectonics*, 6(6), 687-705.
- Khalid A. Almalki, Peter G. Betts, Ailleres L. (2015). The Red Sea – 50 years of geological and geophysical research. *Earth Science Review* (147), 109-140..
- Lalou, C., & Hoang, C. T. (1979). Les méthodes de datation par les descendants de l'Uranium. *Bulletin de l'Association française pour l'étude du quaternaire*, 16(1), 3-14.
- Manaa, A. A., Jones, B. G., McGregor, H. V., Zhao, J.-X., and Price, D. M. (2016). Dating quaternary raised coral terraces along the Saudi Arabian Red Sea coast. *Mar. Geol.* 374, 59–72.
- Pautot, G. (1983). Les fosses de la Mer Rouge: approche géomorphologique d'un stade initial d'ouverture océanique réalisée à l'aide du Seabeam. *Oceanologica acta*, 6(3), 235-244.
- Raitso, D. E., Pradhan, Y., Brewin, R. J., Stenchikov, G., & Hoteit, I. (2013). Remote sensing the phytoplankton seasonal succession of the Red Sea. *PLoS One*, 8(6), e64909.